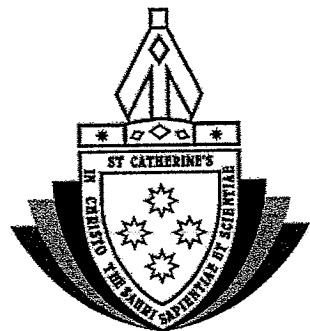




St. Catherine's School
Waverley

August 2008

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Extension I Mathematics

Time allowed: 2 Hours + 5 mins Reading Time

INSTRUCTIONS

- Write your STUDENT NUMBER on each page
- All questions are of equal value
- Marks for each part of a question are indicated
- All questions should be attempted on the separate paper provided
- All necessary working should be shown
- Start each question on a NEW page
- Approved scientific calculators and drawing templates may be used
- Standard integrals are printed at the end of the paper

Student Number: _____

QUESTION 1 (12 marks)

Marks

a) Solve the inequality $\frac{2x+5}{x-4} \leq 1$

3

b) Evaluate $\int_{-3}^3 \frac{1}{9+x^2} dx$

2

c) Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{5x}$

2

d) Given that $\log_b \left(\frac{p}{q} \right) = 3$ and $\log_b \left(\frac{q}{r} \right) = 1.6$, evaluate $\log_b \left(\frac{p}{r} \right)$

2

e) Evaluate $\int_0^{\frac{1}{2}} 2x\sqrt{1-2x} dx$ using the substitution $u = 1-2x$

3

QUESTION 2 (12 marks) Start a new page.

Marks

a) $\tan \theta = m$ and $\tan \phi = 3$ find the value of m if $\theta - \phi = \frac{\pi}{4}$

2

b) Prove that, if $x^4 - x^3 + kx - 4$ has a factor of $(x+1)$, then it also has a factor of $(x-2)$.

2

c) Prove that $\frac{2}{\cot x + \tan x} = \sin 2x$

2

d) Find the general solution of $\sqrt{3} \sin 2x = \cos 2x$

2

e) Consider the function $f(x) = \frac{\pi}{2} + 2 \sin^{-1} \left(\frac{2x}{3} \right)$

(i) Find the domain and range of $f(x)$

2

(ii) Sketch the graph of $f(x)$ showing clearly its end points.

2

QUESTION 3 (12 marks) Start a new page.

Marks

a) Use mathematical induction to show that, for all positive integers $n \geq 1$,
 $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$

3

b) Find the value of the term that is independent of x in the expansion of
 $\left(2x^2 + \frac{1}{x^3} \right)^{10}$

2

c) $Q(x) = ax^2 + bx + c$

(i) State the sum of the roots of $Q(x) = 0$

1

(ii) When $Q(x)$ is divided by either $(x-m)$ or $(x-n)$ the remainder is the same.
 Prove that, if $m \neq n$, then $(m+n)$ is equal to the sum of the roots of $Q(x) = 0$

2

d) Consider the function $f(x) = \frac{x-2}{x-1}$.

(i) Prove that $f(x)$ is an increasing function for all values of x .

2

(ii) Find the equation of the inverse function $f^{-1}(x)$ and deduce that $f(x)$ is symmetrical about the line $y = x$

2

QUESTION 4 (12 marks) Start a new page.

Marks

a) (i) Prove that $\int_0^{\frac{\pi}{4}} \sin^2 x \, dx = \frac{\pi}{8} - \frac{1}{4}$

2

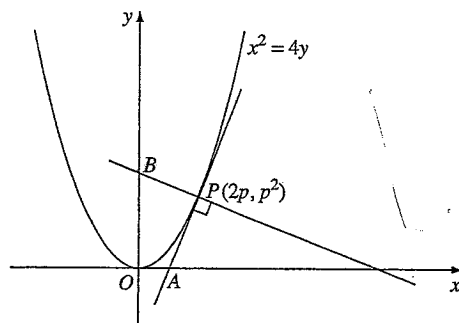
(ii) Prove that $\frac{d}{dx}(x \sin^2 x) - \sin^2 x = x \sin 2x$

2

(iii) Hence or otherwise, prove $\int_0^{\frac{\pi}{4}} x \sin 2x \, dx = \frac{1}{4}$

2

b)



6

The diagram shows the graph of $x^2 = 4y$. The tangent to the parabola at $P(2p, p^2)$, $p > 0$, cuts the x axis at A . The normal to the parabola at P cuts the y axis at B .

(i) Derive the equation of the tangent AP

2

(ii) Show that B has coordinates $(0, p^2 + 2)$.

1

(iii) Let C be the midpoint of AB . Find the Cartesian equation of the locus of C .

3

QUESTION 5 (12 marks) Start a new page.

Marks

a) The rate at which a drug is being expelled from the body at time t hours is given by the equation $\frac{dM}{dt} = -k(M - 0.04)$ where k is a constant and M is measured in grams.

6

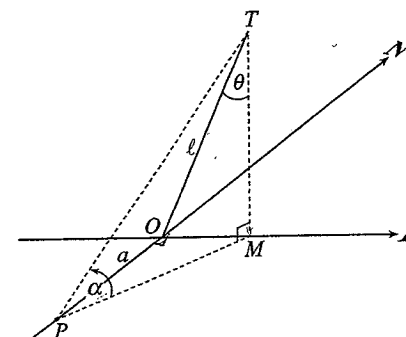
(i) Show that $M = 0.04 + M_0 e^{-kt}$, for some constant M_0 , satisfies this equation.

(ii) Initially 4 grams was ingested. Find the value of M_0

(iii) After 10 hours, 1.6 grams was still present. Find the value of k

(iv) Show that the drug will never be entirely eliminated from the body.

b)



A pole, OT , of length ℓ m, stands on horizontal ground. The pole leans towards the east, making an angle of θ with the vertical. From P , a m south of O the elevation of T is α .

(i) Find expressions, in terms of ℓ and θ , for OM and MT

(ii) Prove that $PM = \ell \cos \theta \cot \alpha$.

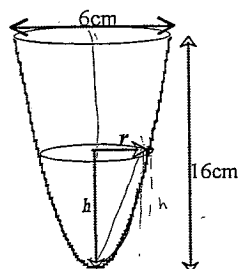
(iii) Prove that $\ell^2 = \frac{a^2}{\cos^2 \theta \cot^2 \alpha - \sin^2 \theta}$

(iv) Find the length of the pole, to the nearest m, if $a = 25$, $\theta = 20^\circ$ and $\alpha = 24^\circ$

QUESTION 6 (12 marks) Start a new page.

Marks

- a) A wine glass is formed by rotating $y = ax^2$ around the y axis.



The depth of liquid in the glass is h and the radius at the top of the liquid is r .

- (i) Find the value of a 1
- (ii) Write an expression for h in terms of r . 1
- (iii) Show that the volume of liquid in the glass 1
when the depth is h cm is $\frac{8\pi r^4}{9}$
- (iv) Liquid is being added to the glass at the 3
at a rate $3(15 - h)$ ml per second. Find the rate
at which the radius of the surface is increasing
when $h = 10$ cm.

- b) A particle moves in a straight line such that its displacement from a fixed point O 6
is given by ;

$$x = \sqrt{3} \cos 3t - \sin 3t$$

- (i) Show that $\ddot{x} = -9x$
- (ii) Express x in the form $a \cos(nt + \alpha)$ and hence determine the period and amplitude of the motion.
- (iii) Find the speed of the particle when it is 1 m from O .
- (iv) After how many seconds will the particle be 1 m from O

QUESTION 7 (12 marks) Start a new page.

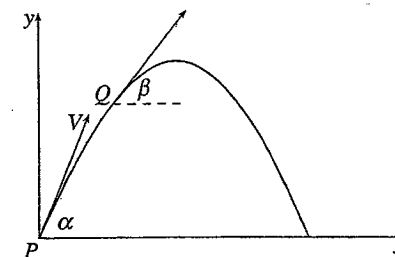
Marks

- a) (i) Show that, in the binomial expansion of $\left(x - \frac{1}{x}\right)^{2n}$, the term independent of x is $(-1)^n {}^{2n}C_n$ 2

- (ii) Show that $(1+x)^{2n} \left(1 - \frac{1}{x}\right)^{2n} \equiv \left(x - \frac{1}{x}\right)^{2n}$ 2

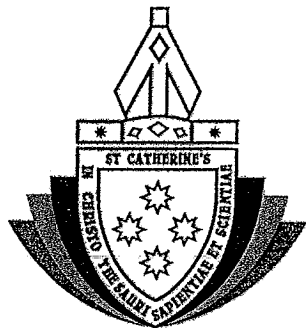
- (iii) Deduce that 2
$$({}^{2n}C_0)^2 - ({}^{2n}C_1)^2 + ({}^{2n}C_2)^2 - \dots + ({}^{2n}C_{2n})^2 = (-1)^n {}^{2n}C_n$$

- b) A particle is projected from a point P on horizontal ground, with initial speed V metres per second at an angle of elevation to the horizontal of α .



Its equations of motion are $\ddot{x} = 0$, $\ddot{y} = -g$

- (i) Derive expressions for its horizontal and vertical displacements from P after t seconds 2
- (ii) Determine the time of flight of the particle 2
- (iii) The particle reaches the point Q , as shown, where the direction of the flight makes an angle of β with the horizontal. Show that the time taken to travel from P to Q is $\frac{V \sin(\alpha - \beta)}{g \cos \beta}$ seconds 2



St. Catherine's School
Waverley

August 2008

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Extension 1 Mathematics Solutions

Time allowed: 2 Hours + 5 mins Reading Time

INSTRUCTIONS

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Student Number: _____

Course: MATHEMATICS EXTENSION 1

Marking Scheme for Task: HSC TRIAL EXAMINATION

Academic Year: 2007-8

Solutions	Marks	Comments
<u>Question 1:</u> a) $\frac{2x+5}{x-4} \leq 1 \quad (x \neq 4)$ $(x-4)(2x+5) \leq (x-4)^2$ $2x^2 - 3x - 20 \leq x^2 - 8x + 16$ $x^2 + 5x - 36 \leq 0$ $(x+9)(x-4) \leq 0$ $-9 \leq x \leq 4$ but $x \neq 4 \quad \therefore -9 \leq x < 4$ b) $\int_{-3}^3 \frac{1}{9+x^2} dx = 2 \int_0^3 \frac{1}{9+x^2} dx \quad (\text{even function})$ $= 2 \cdot \frac{1}{3} \left[\tan^{-1} \frac{x}{3} \right]_0^3$ $= \frac{2}{3} \left[\frac{\pi}{4} \right]$ $= \frac{\pi}{6}$ c) $\lim_{x \rightarrow 0} \frac{\sin x}{5x} = \frac{1}{5} \lim_{x \rightarrow 0} \frac{\sin x}{x}$ $= \frac{1}{5}$ d) $\log_e \left(\frac{p}{q} \right) = \log \left(\frac{p}{q} \times \frac{q}{r} \right) \quad \text{line 1}$ $= \log \frac{p}{q} + \log \frac{q}{r} \quad \text{line 2}$ $= 3 + 1.6 \quad \text{line 3}$ $= 4.6$ e) $\int_0^{1/2} 2x \sqrt{1-2x} dx$ $u = 1-2x \quad \therefore 2x = 1-u$ $du = -2dx \quad \therefore dx = -\frac{du}{2}$ $x=0 \quad u=1$ $x=\frac{1}{2} \quad u=0$	3 2 2 2	1 for arriving at this line 1 for factorising + sketch 1 correct answer * (-0.5 if $-9 \leq x \leq 4$) 1 correct primitive 1 correct answer 1 correct simplifying $\frac{1}{2} \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ 1 correct answer 1 mark for line 1 0.5 mark for line 2 0.5 (line 3)

Solutions

Marks

Comments

e) (continued)

$$= - \int_1^0 (1-u) \sqrt{u} \cdot \frac{du}{2} \quad \text{line 1}$$

$$= - \frac{1}{2} \int_1^0 (u^{1/2} - u^{3/2}) du \quad \text{line 2}$$

$$= - \frac{1}{2} \left[\frac{2u^{3/2}}{3} - \frac{2u^{5/2}}{5} \right]_1^0 \quad \text{line 3}$$

$$= - \frac{1}{2} \left[0 - \left(\frac{2}{3} - \frac{2}{5} \right) \right]$$

$$= \frac{2}{15} \quad \text{line 4}$$

1/2 mark for line 1

1/2 mark for line 2

1/2 mark for line 3

1/2 mark for line 4

3 (-0.5 for incorrect limit)

Question 2 a) $\tan \theta = m$ $\tan \phi = 3$

$$\theta - \phi = \frac{\pi}{4}$$

$$\therefore \tan(\theta - \phi) = 1 \quad \left[\tan \frac{\pi}{4} = 1 \right]$$

$$\frac{\tan \theta - \tan \phi}{1 + \tan \theta \tan \phi} = 1 \quad *$$

$$\therefore \frac{m - 3}{1 + 3m} = 1 \quad \square$$

$$m - 3 = 1 + 3m$$

$$-2 = m$$

$$b) f(x) = x^4 - x^3 + kx - 4$$

$$\text{If } x+1 \text{ is a factor } f(-1) = 0$$

$$\therefore 1 + 1 - k - 4 = 0 \Rightarrow k = -2 \quad \text{1 mark for arriving at this line}$$

$$\therefore f(x) = x^4 - x^3 - 2x - 4$$

$$\text{Now } f(2) = 16 - 8 - 4 - 4 = 0$$

$$\therefore (x-2) \text{ is also a factor.}$$

1 mark for * or $\square$

1 mark for solving 'm'

2

1 mark for showing $(x-2)$ is also a factor

2

$$\theta = \tan^{-1} m \text{ and } \phi = \tan^{-1} 3$$

$$\tan^{-1} m - \tan^{-1} 3 = \frac{\pi}{4}$$

$$\tan^{-1} m = \frac{\pi}{4} + \tan^{-1} 3$$

$$m = \tan \left(\frac{\pi}{4} + \tan^{-1} 3 \right)$$

$$m = -2$$

Solutions

Marks

Comments

Question 2 c) LHS = $\frac{2}{\cot x + \tan x} = \frac{2}{\frac{\cos x}{\sin x} + \frac{\sin x}{\cos x}} \quad \text{line 1}$

$$= \frac{2 \sin x \cos x}{\cos^2 x + \sin^2 x} \quad \text{line 2}$$

$$= \sin 2x \quad \text{line 3}$$

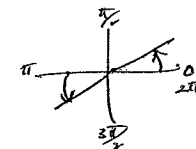
$$d) \sqrt{3} \sin 2x = \cos 2x$$

$$\tan 2x = \frac{1}{\sqrt{3}} \quad \times$$

$$2x = n\pi + \tan^{-1} \frac{1}{\sqrt{3}}$$

$$2x = n\pi + \frac{\pi}{6}$$

$$\therefore x = \frac{n\pi}{2} + \frac{\pi}{12} \quad n = 0, 1, 2, \dots$$



1 mark for line 1
1 mark for line 2
1/2 mark for line 3

2

'x' means 1/2 marks

$$e) f(x) = \frac{\pi}{2} + 2 \sin^{-1} \frac{2x}{3}$$

$$\text{Domain: } -1 \leq \frac{2x}{3} \leq 1$$

$$-3 \leq 2x \leq 3$$

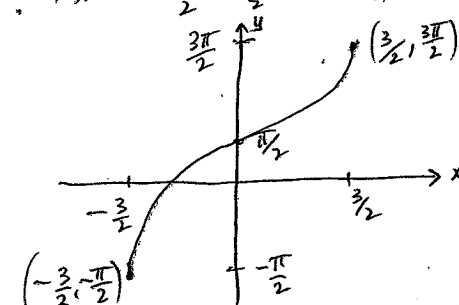
$$-\frac{3}{2} \leq x \leq \frac{3}{2} \quad \checkmark$$

$$\text{Range: } -\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2} \quad y = \frac{\pi}{2}$$

$$-\pi \leq 2 \sin^{-1} x \leq \pi$$

$$-\frac{\pi}{2} \leq \frac{\pi}{2} + 2 \sin^{-1} x \leq \frac{3\pi}{2}$$

$$\therefore -\frac{\pi}{2} \leq \frac{\pi}{2} + 2 \sin^{-1} \left(\frac{2x}{3} \right) \leq \frac{3\pi}{2} \quad \checkmark$$



'✓' means 1 mark.

2

1 for curve
1 for clearly showing help points

2

1/2 for correct y-intercept

Course:

Page no. 4 of 12

Marking Scheme for Task:

Academic Year: 2007-8

Solutions

Marks

Comments

Question 3 a) $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$

① for $n=1$ LHS = $1^2 = 1$ RHS = $\frac{1}{3} \cdot 1 \cdot (1)(3) = 1$
 $\therefore$ true for $n=1$

② assume true for $n=k$

i.e. $1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 = \frac{1}{3}k(2k-1)(2k+1)$

③ Aim to prove true for $n=k+1$ if true for $n=k$

i.e. AIP. $1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 + (2k+1)^2 = \frac{1}{3}(k+1)(2k+1)(2k+3)$

$$\begin{aligned} \text{LHS} &= \frac{1}{3}k(2k-1)(2k+1) + (2k+1)^2 \\ &= (2k+1) \left[\frac{1}{3}k(2k-1) + (2k+1) \right] \\ &= (2k+1) \left[\frac{2k^2 - k}{3} + 2k+1 \right] \\ &= (2k+1) \left(\frac{2k^2 - k + 6k + 3}{3} \right) \\ &= \frac{1}{3}(2k+1)(2k^2 + 5k + 3) \\ &= \frac{1}{3}(2k+1)(2k+3)(k+1) = \text{RHS} \end{aligned}$$

$\therefore$ true for $n=k+1$ if true for $n=k$

Since true for $n=1$ then by theory of mathematical induction true for all $n \geq 1$

b) $(2x + \frac{1}{x^3})^{10}$ $T_{k+1} = {}^{10}C_k (2x^2)^{10-k} (\frac{1}{x^3})^k$

$x^{20-2k-3k} = x^0 \Rightarrow 20-5k=0 \therefore k=4 \checkmark$

$T_5 = {}^{10}C_4 (2x^2)^6 (\frac{1}{x^3})^4 = 210 \cdot 64 \cdot x^{12} \cdot \frac{1}{x^{12}}$
 $= 13440 \checkmark$ or ${}^{10}C_4 2^6$

$\frac{1}{2}$ mark for ①
 $\frac{1}{2}$ mark for ②

2 marks for ③

3

$\checkmark$ means 1 mark
 $\times$ means 0.5 mark

2

Course:

Page no. 5 of 12

Marking Scheme for Task:

Academic Year: 2007-8

Solutions

Marks

Comments

Question 3, c) $Q(x) = ax^2 + bx + c$

(i) sum of roots = $-\frac{b}{a}$

(ii) remainder when $Q(x)$ divided by $(x-m)$

$Q(m) = am^2 + bm + c \checkmark$

remainder when $Q(x)$ divided by $(x-n)$

$Q(n) = an^2 + bn + c \checkmark$

now $am^2 + bm + c = an^2 + bn + c$

$am^2 - an^2 = -bm + bn \checkmark$

$a(m-n)(m+n) = -b(m-n) \checkmark$

$\therefore m+n = -\frac{b}{a} = \text{sum of roots}$
 from part (i)

d) $f(x) = \frac{x-2}{x-1}$

(i) $f'(x) = \frac{(x-1) - (x-2)}{(x-1)^2}$

$= \frac{1}{(x-1)^2} \geq 0$ for all $x, x \neq 1$

$\therefore f(x)$ is increasing for all x

(ii) let $y = \frac{x-2}{x-1}$

$\therefore x = \frac{y-2}{y-1}$ is inverse

$xy - x = y - 2$

$xy - y = x - 2$

$y(x-1) = x-2$

$\therefore y = \frac{x-2}{x-1}$ is inverse function

$\therefore f(x)$ is its own inverse

$\therefore f(x)$ is symmetrical about $y=x$

$f(x)$ is invertible (1 mark)

$f'(x) > 0$
 (1 mark)

$\frac{1}{(x-1)^2} > 0$
 (1 mark)

1 for finding inverse

1 mark.

Course:

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Marking Scheme for Task:

Academic Year: 2007-8

Solutions	Marks	Comments
<u>Question 4:</u> a) (i) $\int_0^{\pi/4} \sin^2 x \, dx = \int_0^{\pi/4} \frac{1}{2}(1 - \cos 2x) \, dx$ ✓ $= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi/4}$ $= \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{2} \right] \checkmark$ $= \frac{\pi}{8} - \frac{1}{4}$	2	
(ii) $\frac{d}{dx}(x \sin^2 x) = \sin^2 x$ ✓ $= \sin^2 x + x \cdot 2 \sin x \cos x = \sin^2 x + x \sin 2x \checkmark$	2	
(iii) $\int_0^{\pi/4} x \sin 2x \, dx$ $= \left[x \sin^2 x \right]_0^{\pi/4} - \int_0^{\pi/4} \sin^2 x \, dx \checkmark$ $= \frac{\pi}{8} - \left(\frac{\pi}{8} - \frac{1}{4} \right) \text{ from part (i)} \checkmark$ $= \frac{1}{4}$	2	
b) (i) $P(2p, p^2)$ $y = \frac{x^2}{4}$ ✓ $y' = \frac{x}{2} \checkmark$ at P $y' = p$ ✓ $y - p^2 = p(x - 2p) \checkmark$ $y - p^2 = px - 2p^2 \checkmark$ $y - px + p^2 = 0 \text{ is tangent AP.} \checkmark$	2	

Course:

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Marking Scheme for Task:

Academic Year: 2007-8

Solutions	Marks	Comments
b) (ii) normal: $y - p^2 = -\frac{1}{p}(x - 2p)$ ✓ $py - p^3 = -x + 2p \checkmark$ let $x=0$ $py - p^3 = 2p$ ✓ $py = 2p + p^3$ $y = 2 + p^2 \checkmark$ $\therefore B(0, 2+p^2)$ (iii) $A(p, 0)$ $B(0, p^2+2)$ $\therefore C\left(\frac{p}{2}, \frac{p^2+2}{2}\right) \checkmark$ ① $x = \frac{p}{2}$ ② $y = \frac{p^2+2}{2}$ from ① $p = 2x$ ✓ in ② $y = \frac{4x^2+2}{2} \checkmark$ $\therefore y = 2x^2 + 1$ is locus of C ✓	2	
<u>Question 5:</u> a) (i) $M = 0.04 + M_0 e^{-kt}$ — ① $\frac{dM}{dt} = -k M_0 e^{-kt}$ but from ① $M_0 e^{-kt} = M - 0.04$ $\therefore \frac{dM}{dt} = -k(M - 0.04)$	2	
(ii) when $t=0$ $M=4$ $\therefore 4 = 0.04 + M_0 \Rightarrow M_0 = 3.96$	1	
(iii) when $t=10$ $M=1.6$ $\therefore M = 1.6 = 0.04 + 3.96 e^{-10k}$ $1.56 = 3.96 e^{-10k}$ $e^{-10k} = \frac{1.56}{3.96}$	2	full marks if wrong M_0 used

Course:

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Marking Scheme for Task:

Academic Year: 2007-8

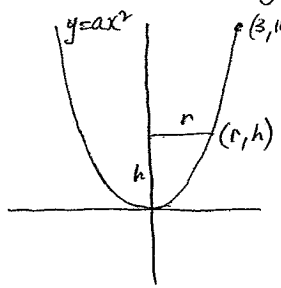
Solutions	Marks	Comments
<u>Question 5 (continued)</u> $\therefore -10k = \ln\left(\frac{1.56}{3.96}\right)$ $k = \frac{\ln\left(\frac{1.56}{3.96}\right)}{-10}$ $= 0.0932 \text{ (4 d.p.)}$ (iv) $M = 0.04 + 3.96e^{-0.0932t}$ as $t \rightarrow \infty$ $e^{-0.0932t} \rightarrow 0$ $\therefore M \rightarrow 0.04 \therefore M \neq 0$ thus never eliminated. b). (i) $OM = l \sin \theta$ $MT = l \cos \theta$ (ii) $\tan \alpha = \frac{MT}{PM}$ $\therefore PM = \frac{MT}{\tan \alpha}$ $= l \cos \theta \cdot \frac{1}{\tan \alpha}$ $= l \cos \theta \cot \alpha$ (iii) in $\triangle POM$ $PM^2 - OM^2 = a^2$ (pythagoras) $\therefore l^2 \cos^2 \theta \cot^2 \alpha - l^2 \sin^2 \theta = a^2$ $l^2 (\cos^2 \theta \cot^2 \alpha - \sin^2 \theta) = a^2$ $\therefore l^2 = \frac{a^2}{\cos^2 \theta \cot^2 \alpha - \sin^2 \theta}$ (iv) $l^2 = \frac{25^2}{\cos^2 20^\circ \cot^2 24^\circ - \sin^2 20^\circ}$ $\therefore l \div 12$	1 1 1 2 2 1	1/2 for ratio 1/2 each 1/2 for ratio

Course:

Page no. 9 of 12

Marking Scheme for Task:

Academic Year: 2007-8

Solutions	Marks	Comments
<u>Question 6</u> $y = ax^2$  (i) curve passes through (3, 16) $\therefore 16 = 9a$ $a = \frac{16}{9}$ (ii) curve passes through (r, h) $\therefore h = ar^2$ $h = \frac{16}{9}r^2$ (iii) $V = \pi \int_a^b x^2 dy$ note $y = \frac{16}{9}x^2$ $\therefore x^2 = \frac{9y}{16}$ $= \pi \int_0^h \frac{9y}{16} dy$ $= \pi \left[\frac{9y^2}{32} \right]_0^h$ $= \frac{9\pi h^2}{32}$ but $h = \frac{16}{9}r^2$ $\therefore h^2 = \frac{16^2}{9^2}r^4$ $\therefore V = \frac{9\pi}{32} \times \frac{16^2}{9^2}r^4$ $= \frac{8\pi r^4}{9}$ (iv) $\frac{dv}{dt} = 3(15-h)$ find $\frac{dr}{dt}$ $\frac{dr}{dt} = \frac{dv}{dt} \times \frac{dr}{dv}$ $= 15 \times \frac{1}{15\sqrt{10}\pi}$ $= \frac{1}{\sqrt{10}\pi} \times \frac{\sqrt{10}}{\sqrt{10}}$ $= \frac{\sqrt{10}}{10\pi} \text{ cm/s}$ $V = \frac{8\pi r^4}{9}$ $\therefore \frac{dv}{dr} = \frac{32\pi r^3}{9}$ $= \frac{32}{9} \times \pi \times \left(\frac{3}{4}\sqrt{h}\right)^3$ $= \frac{3\pi 10\sqrt{10}}{2}$ $= 15\sqrt{10}\pi$ when $h = 10$	1 1 2 2	1/2 for $\frac{dv}{dt}$ 2 from (ii)

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Solutions

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Comments

Question 6 b) (i)

$$x = \sqrt{3} \cos 3t - \sin 3t$$

$$\dot{x} = -3\sqrt{3} \sin 3t - 3 \cos 3t$$

$$\ddot{x} = -9\sqrt{3} \cos 3t + 9 \sin 3t$$

$$= -9(\sqrt{3} \cos 3t - \sin 3t)$$

$$\therefore \ddot{x} = -9x \quad (\text{S.H.M.})$$

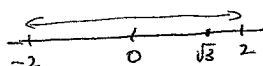
$$(ii) \quad x = 2 \cos\left(3t + \frac{\pi}{6}\right)$$



amplitude = 2

$$\text{period} = \frac{2\pi}{n} = \frac{2\pi}{3}$$

Note: when $t=0$ $x = \sqrt{3}$
 $x=0$ $\ddot{x}=0$ $\therefore$ centre origin



$$(iii) \text{ When } x=1 \quad \cos\left(3t + \frac{\pi}{6}\right) = \frac{1}{2}$$

$$3t + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \dots$$

$$3t = \frac{\pi}{6}, \frac{9\pi}{6}, \dots$$

$$t = \frac{\pi}{18}, \frac{\pi}{2}, \dots$$

$$\text{now speed} = |-3\sqrt{3} \sin 3t - 3 \cos 3t|$$

$$\text{When } t = \frac{\pi}{18} \quad = \left| -3\sqrt{3} \cdot \frac{1}{2} - \frac{3\sqrt{3}}{2} \right|$$

$$= \left| -\frac{3\sqrt{3}}{2} - \frac{3\sqrt{3}}{2} \right|$$

$$= 3\sqrt{3} \text{ m/s}$$

$$(iv) \text{ Particle is at } x=1 \text{ when } t = \frac{\pi}{18} \text{ secs}$$

then $\frac{\pi}{2}$ secs

1

2

2

1

Or use

$$V^2 = n^2(a^2 - x^2)$$

Solutions

Marks

Comments

Question 7 (i)

$$\left(x - \frac{1}{x}\right)^{2n} \quad T_{k+1} = {}^{2n}C_k x^{2n-k} \left(-\frac{1}{x}\right)^k$$

$$= {}^{2n}C_k x^{2n-k-k} (-1)^k$$

$$\text{for term independent of } x \quad 2n - 2k = 0$$

$$\therefore k = n$$

$$\therefore T_{n+1} = {}^{2n}C_n (-1)^n$$

$$(ii) \quad (1+x)^{2n} \left(1 - \frac{1}{x}\right)^{2n} = \left[(1+x)\left(1 - \frac{1}{x}\right)\right]^{2n}$$

$$= \left(1 - \frac{1}{x} + x - 1\right)^{2n}$$

$$= \left(x - \frac{1}{x}\right)^{2n}$$

(ii) examine terms independent of x

$$\text{LHS} = \left[{}^{2n}C_0 + {}^{2n}C_1 x + {}^{2n}C_2 x^2 + \dots + {}^{2n}C_{2n} x^{2n} \right] \left[{}^{2n}C_0 - {}^{2n}C_1 \left(\frac{1}{x}\right) + {}^{2n}C_2 \left(\frac{1}{x}\right)^2 - \dots + {}^{2n}C_{2n} \left(\frac{1}{x}\right)^{2n} \right]$$

terms independent of x are

$${}^{2n}C_0 {}^{2n}C_0 - {}^{2n}C_1 {}^{2n}C_1 + {}^{2n}C_2 {}^{2n}C_2 - \dots - {}^{2n}C_{2n} {}^{2n}C_{2n}$$

RHS term independent of x is $(-1)^n {}^{2n}C_n$ from (i)

$$\therefore {}^{2n}C_0 - {}^{2n}C_1 + {}^{2n}C_2 - \dots + {}^{2n}C_{2n} = (-1)^n {}^{2n}C_n$$

2

must mention
independent
of x terms
for 1 mark

Solutions

Marks

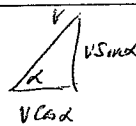
Comments

Q7 d-(i) $\ddot{x} = 0$ $\ddot{y} = -g$

$\dot{x} = c$ $\dot{y} = -gt + c$

when $t=0$ $\dot{x} = V \cos \alpha$ when $t=0$ $\dot{y} = V \sin \alpha$

$\therefore c = V \cos \alpha$



$\dot{x} = V \cos \alpha$ $\therefore \dot{y} = -gt + V \sin \alpha$

$x = Vt \cos \alpha + c$ $y = -\frac{gt^2}{2} + Vt \sin \alpha + c$

when $t=0$ $x=0 \therefore c=0$ when $t=0$ $y=0 \therefore c=0$

$\therefore x = Vt \cos \alpha$ $y = Vt \sin \alpha - \frac{1}{2}gt^2$

2

2

(ii) at time of flight $y=0$

$\therefore Vt \sin \alpha - \frac{1}{2}gt^2 = 0$

$2Vt \sin \alpha - gt^2 = 0$

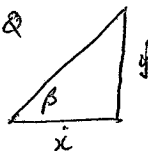
$t(2V \sin \alpha - gt) = 0$

$\therefore t = 0, \frac{2V \sin \alpha}{g}$

$\therefore \text{time of flight} = \frac{2V \sin \alpha}{g}$

2

(iii) at Q



$\tan \beta = \frac{y}{x}$

$\frac{\sin \beta}{\cos \beta} = \frac{V \sin \alpha - gt}{V \cos \alpha}$

$V \sin \beta \cos \alpha = V \sin \alpha \cos \beta - gt \cos \beta$

$gt \cos \beta = V(\sin \alpha \cos \beta - \cos \alpha \sin \beta)$

$\therefore t = \frac{V \sin(\alpha - \beta)}{g \cos \beta}$ See Q 2